

Targeted therapies for breast cancer

Targeted therapy drugs can reduce the risk of breast cancer coming back by affecting the way cancer cells grow.

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What are targeted therapies?

[Targeted therapies](#) interfere with the way cancer cells grow. The targeted therapy drugs used to treat breast cancer are [trastuzumab](#) (often called Herceptin[®]) and pertuzumab (Perjeta[®]). They help to reduce the risk of [HER2 positive breast cancer](#) coming back.

You may have trastuzumab on its own or with pertuzumab. They may be given together as an injection called [Phesgo](#)[®]. Trastuzumab and pertuzumab attach to the HER2 receptors on the breast cancer cells and stop them from dividing and growing.

Trastuzumab and pertuzumab may also be given with chemotherapy. You may have them before or after surgery and radiotherapy.

There are other types of targeted therapy drug to treat HER positive breast cancer. You may have these if you have already had trastuzumab and pertuzumab.

Trastuzumab

You usually have [trastuzumab](#) every 3 weeks for 1 year. Sometimes you may have it over 6 months. Your cancer doctor will tell you how long you need to have it for. You have it in the chemotherapy day unit or outpatient department.

You usually have trastuzumab as an injection under the skin (subcutaneously). But you may have it as a drip into a vein (an infusion).

Side effects of trastuzumab

The side effects of trastuzumab are usually mild. You may have some side effects while you are having the infusion or injection, or up to 4 hours after. This is more likely to happen with the first dose. These side effects include flu-like symptoms, such as:

- a headache
- a high temperature (fever) and chills
- feeling sick.

Any side effects usually get better a few hours after you finish the treatment.

Another possible side effect is an allergic reaction, but this is rare. The nurses will check for signs of a reaction. If it happens, they can treat it quickly with drugs.

You may get other side effects after treatment. These include:

- diarrhoea
- headaches
- feeling sick.

Effects on the heart

Trastuzumab may cause changes in the way the heart works. This can cause problems in some people. Usually, the effect is mild and returns to normal after treatment ends. You may be given heart medicines to help with this side effect.

You usually have tests to check your heart before and during treatment. This is usually a scan of the heart called an echocardiogram. It helps to make sure the drug is not causing any damage.

Trastuzumab is not usually given to people who have serious heart problems. If there are any changes with your heart, you may be referred to a heart specialist doctor called a cardiologist. They may give you drugs to support your heart.

Pertuzumab

You have pertuzumab every 3 weeks with trastuzumab. You have it as an infusion. The first dose is usually given slowly over about 60 minutes. This is because some people can have a reaction.

The nurses monitor you during the infusion and for about 60 minutes afterwards. If you have no problems, you can have the next doses over 30 to 60 minutes.

Side effects of pertuzumab

The side effects of pertuzumab include:

- a headache
- diarrhoea, which can be severe
- feeling sick
-

itchy skin or a rash.

Pertuzumab can also affect the heart. But you will have regular heart checks during treatment. Your cancer doctor or a nurse will give you more information about the side effects and ways of coping with them.

Phesgo

[Phesgo[®]](#) is a combination of trastuzumab and pertuzumab. You have it as an injection under the skin (subcutaneously). The nurse gives you the injection every 3 weeks, into the thigh.

Other types of targeted therapy drug

Trastuzumab emtansine

[Trastuzumab emtansine](#) is an [antibody drug conjugate](#). It is trastuzumab with the chemotherapy drug emtansine attached to it. You have it as an infusion every 3 weeks.

You may be offered trastuzumab emtansine if cancer cells are still in the breast or [lymph nodes](#) after surgery. It is offered to people who have had [chemotherapy](#) and HER2 treatment before surgery ([neo-adjuvant treatment](#)).

Neratinib

You take [neratinib](#) as tablets. It may be used to treat breast cancer that is both [HER2 positive and ER positive](#). It may be offered if you have had surgery and treatment with trastuzumab less than 1 year ago. Your cancer doctor or breast care nurse can explain more about whether this drug is suitable for you.

Abemaciclib

You take [abemaciclib](#) as tablets. It may be used to treat early breast cancer that has a higher risk of coming back.

Olaparib

You take [olaparib](#) as tablets. It may be used to treat early breast cancer that is HER2 negative and has a higher risk of coming back. It is given to people who have a gene change called [BRCA1 or BRCA2](#). Your cancer doctor or nurse will explain if this drug is suitable for you.

Triple negative breast cancer

If you have triple negative breast cancer, you may be offered other types of targeted therapy drug. We have more information about these on our page on [triple negative breast cancer](#).

About our information

References



Below is a sample of the sources used in our breast cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

ESMO. Early breast cancer clinical practice guidelines for diagnosis, treatment and follow-up. 2019, Vol 30, pp1192–1220. Available from: <https://www.esmo.org/guidelines/guidelines-by-topic/breast-cancer/early-breast-cancer> [accessed 2023].

National Institute for Health and Care Excellence (NICE). Early and locally advanced breast cancer: diagnosis and management. 2018. Updated 2023. Available from: <https://www.nice.org.uk/guidance/ng101> [accessed 2023].

Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by [Dr Rebecca Roylance](#), Consultant Medical Oncologist and [Professor Mike Dixon](#), Professor of Surgery and Consultant Breast Surgeon.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
-

explain medical words

- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as 'you' so that everyone feels included. Where clinically necessary we use the terms 'men' and 'women' or 'male' and 'female'. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information [here](#).

Date reviewed

Reviewed: 01 October 2023 | Next review: 01 October 2026

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community



Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask questions.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help